Examiner only

8. (a) Chlorine, bromine and iodine are Group 7 elements.

(i) Describe and explain the trend in reactivity within the group. [2]

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(ii) **X**, **Y** and **Z** are known to be chlorine, bromine and iodine but not necessarily in that order.

A student identifies **X**, **Y** and **Z** by mixing them with solutions of halide salts.

The table shows the observations made.

Element	Sodium bromide	Sodium iodide	Sodium chloride
X	solution turns orange	solution turns brown	no change
Y	no change	no change	no change
Z	no change	solution turns brown	no change

Use the observations to identify elements **X**, **Y** and **Z**.

Explain your reasoning. [2]

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(b) Iron reacts with chlorine gas to form iron(III) chloride according to the equation shown.

$$2Fe + 3Cl_2 \longrightarrow 2FeCl_3$$

Calculate the mass of iron(III) chloride that could be produced using 22.4 g of iron. [2]

$A_r(Fe) = 56$ $A_r(Cl) = 35.5$

Mass = ... g

(c) White phosphorus has the formula P_4. It reacts with chlorine to form phosphorus trichloride, PCl_3.

Put a number in each box to balance the equation for this reaction. [1]

$$P_4 + \boxed{}\,Cl_2 \longrightarrow \boxed{}\,PCl_3$$

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